

Title: Mobile and Wireless Computing Exam

Course Code: IT455

Instructor: Dr. Shatha

Duration: 45

Q1. What is the primary purpose of a WLAN? (mcq) [1]

- a) To replace wired backbone networks
- b) To provide wireless access and replace office cabling
- c) To increase wired network speed
- d) To provide internet access only

Correct Answer: B

Q2. Which frequency band is commonly shared between WLANs and other devices like phones? (mcq) [1]

- a) 5 GHz
- b) 900 MHz
- c) 2.4 GHz
- d) 60 GHz

Correct Answer: C

Q3. In Mobile IP, which node intercepts packets destined for the Mobile Node and tunnels them? (mcq) [1]

- a) Foreign Agent
- b) Correspondent Node
- c) Home Agent
- d) Access Point

Correct Answer: C

Q4. What does CoA stand for in Mobile IPv4? (mcq) [1]

- a) Core of Access
- b) Care-of-Address
- c) Connection over Air
- d) Central Originating Address

Correct Answer: B

Q5. What is the role of the Foreign Agent in Mobile IP? (mcq) [1]

- a) Assigns home addresses to mobile nodes
- b) Decapsulates tunneled packets and forwards them to the mobile node
- c) Manages home network routing tables
- d) Provides DNS resolution for mobile nodes

Correct Answer: B

Q6. Which problem occurs when a wireless station cannot detect collisions due to its own transmissions? (mcq) [1]

- a) Multipath propagation
- b) Path loss
- c) Hidden terminal problem
- d) Carrier sensing problem

Correct Answer: D

Q7. WLANs are typically restricted to small areas such as buildings or campuses. (true_false) [0.5]

- a) True
- b) False

Correct Answer: A

Q8. In Mobile IP, the Mobile Node always sends packets using its foreign IP address. (true_false) [0.5]

- a) True
- b) False

Correct Answer: B

Q9. The Mobility Binding Table is maintained on the Foreign Agent. (true_false) [0.5]

- a) True
- b) False

Correct Answer: B

Q10. Reverse tunneling in Mobile IP solves the ingress filtering problem. (true_false) [0.5]

- a) True
- b) False

Correct Answer: A

Q11. WLANs have a lower Bit Error Rate (BER) compared to wired networks. (true_false) [0.5]

- a) True
- b) False

Correct Answer: B

Q12. MIPv6 requires a Foreign Agent for operation. (true_false) [0.5]

- a) True
- b) False

Correct Answer: B